Tutorial 5: Linear Transformations

22 October 2025

1. Find all matrices $\boldsymbol{X}$ that satisfy each of the following matrix equations.

   (a) $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \boldsymbol{X} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

   (b) $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix} \boldsymbol{X} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

   (c) $\begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \boldsymbol{X} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

   Let us write each matrix equation as $\boldsymbol{AX} = \boldsymbol{B}$. Let $\vec{x}_i$ and $\vec{b}_i$ denote the ith column of $\boldsymbol{X}$ and of $\boldsymbol{B}$, respectively. Observe, then, that

   $$\boldsymbol{A} \begin{pmatrix} \vec{x}_1 & \cdots & \vec{x}_n \end{pmatrix} = \begin{pmatrix} \vec{b}_1 & \cdots & \vec{b}_n \end{pmatrix} \implies \boldsymbol{A}\vec{x}_i = \vec{b}_i \text{ for all } i.$$

   In particular, we can solve each matrix equation by solving a collection of smaller linear systems.

   (a) Since $\boldsymbol{A}$ and $\boldsymbol{B}$ are both 2×2, $\boldsymbol{X}$ must be a 2×2 matrix as well. Let us solve the homogeneous system $\boldsymbol{A}\vec{x} = \vec{0}$:

   $$\left(\begin{array}{cc|c} 1 & 2 & 0 \\ 2 & 4 & 0 \end{array}\right) \xrightarrow{R_2 - 2R_1} \left(\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 0 & 0 \end{array}\right) \implies \vec{x} = \begin{pmatrix} -2t \\ t \end{pmatrix}.$$

   Each column of $\boldsymbol{X}$ must be a vector of this form: hence,

   $$\boldsymbol{X} = \begin{pmatrix} -2s & -2t \\ s & t \end{pmatrix} \text{ for } s, t \in \mathbb{R}.$$

   (b) Since $\boldsymbol{A}$ is 2×3 and $\boldsymbol{B}$ is 2×2, $\boldsymbol{X}$ must be a 3×2 matrix. We row reduce the augmented matrix:

   $$\left(\begin{array}{ccc|c|c} 1 & 2 & 3 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1 \end{array}\right) \xrightarrow{R_1 - 2R_2} \left(\begin{array}{ccc|c|c} 1 & 0 & -1 & 1 & -2 \\ 0 & 1 & 2 & 0 & 1 \end{array}\right).$$

   We thus have solutions to the two linear systems $\boldsymbol{A}\vec{x}_i = \vec{e}_i$: the first column of $\boldsymbol{X}$ is given by the solution to $\boldsymbol{A}\vec{x}_1 = \vec{e}_1$:

   $$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & 0 \end{array}\right) \implies \vec{x} = \begin{pmatrix} 1+s \\ -2s \\ s \end{pmatrix},$$

   and the second column of $\boldsymbol{X}$ is given by the solution to $\boldsymbol{A}\vec{x}_2 = \vec{e}_2$:

   $$\left(\begin{array}{ccc|c} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 1 \end{array}\right) \implies \vec{x} = \begin{pmatrix} -2+t \\ 1-2t \\ t \end{pmatrix}.$$

   Hence, we have

   $$\boldsymbol{X} = \begin{pmatrix} 1+s & -2+t \\ -2s & 1-2t \\ s & t \end{pmatrix} \text{ for } s, t \in \mathbb{R}.$$

(c) We row reduce the augmented matrix:

$$\left(\begin{array}{cc|c|c|c} 1 & 4 & 1 & 0 & 0 \\ 2 & 5 & 0 & 1 & 0 \\ 3 & 6 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3-3R_1]{R_2-2R_1} \left(\begin{array}{cc|c|c|c} 1 & 4 & 1 & 0 & 0 \\ 0 & -3 & -2 & 1 & 0 \\ 0 & -6 & -3 & 0 & 1 \end{array}\right) \cdots$$

$$\xrightarrow{R_3-2R_2} \left(\begin{array}{cc|c|c|c} 1 & 4 & 1 & 0 & 0 \\ 0 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 \end{array}\right).$$

Observe that there is no solution to $\boldsymbol{A}\vec{x}_1 = \vec{e}_1$. Hence, there is no such matrix $\boldsymbol{X}$ that satisfies the given equation.

2. Let the 2×2 matrices $\boldsymbol{A}$ be given by

$$\boldsymbol{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Show that if $ad - bc \neq 0$, then $\boldsymbol{A}$ is invertible and

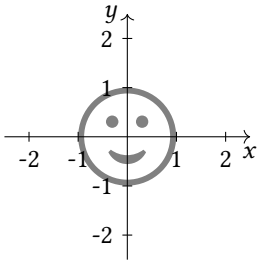
$$\boldsymbol{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

It suffices to show that $\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}_2$:

$$\begin{aligned} \boldsymbol{A}\boldsymbol{A}^{-1} &= \frac{1}{ad - bc} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \\ &= \frac{1}{ad - bc} \begin{pmatrix} ad - bc & -ab + ab \\ cd - cd & -bc + ad \end{pmatrix} = \boldsymbol{I}_2 \end{aligned}$$

$$\begin{aligned} \boldsymbol{A}^{-1}\boldsymbol{A} &= \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ &= \frac{1}{ad - bc} \begin{pmatrix} ad - bc & bd - bd \\ -ac + ac & -bc + ad \end{pmatrix} = \boldsymbol{I}_2. \end{aligned}$$

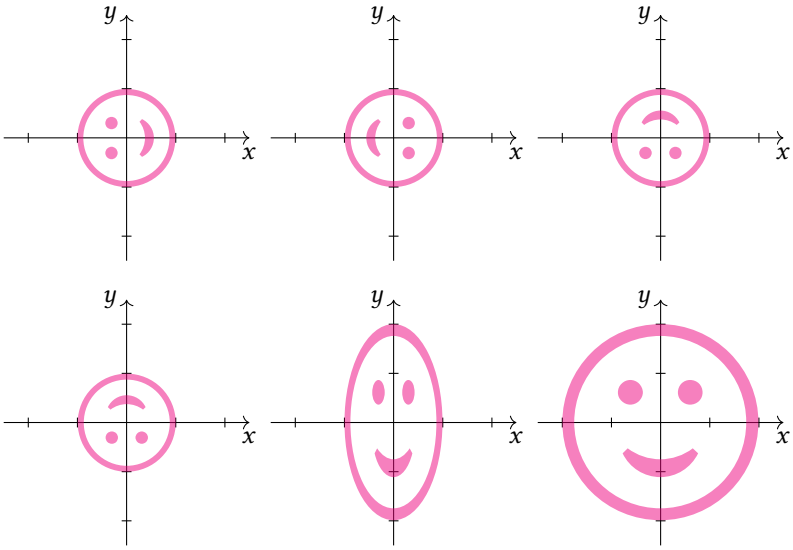
3. Consider the following image on $\mathbb{R}^2$.



For each of the following matrices $\boldsymbol{A}$, draw a sketch showing the effect the transformation $T(\vec{x}) = \boldsymbol{A}\vec{x}$ on the face.

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$



The first matrix represents a 90° counterclockwise rotation.

The second matrix represents a reflection about the line $y = x$.

The third matrix represents a reflection about the x-axis.

The fourth matrix represents a reflection about the origin.

The fifth matrix scales only the y-component by a factor 2.

The sixth matrix represents a scaling by a factor of 2.

4. Let $T(\vec{x}) = \text{proj}_\ell(\vec{x})$ be the transformation corresponding to the projection of $\vec{x} \in \mathbb{R}^2$ onto a line ℓ passing through the origin.

   (a) What is $T(T(\vec{x}))$? Justify your answer.

   (b) Is $T(\vec{x})$ invertible? Justify your answer.

   (a) Since $T(\vec{x}) = \text{proj}_\ell(\vec{x})$ already lies on ℓ, we have $T(T(\vec{x})) = T(\vec{x})$.

   (b) No: multiple vectors may have the same vector as their orthogonal projection onto a line, and so $T(\vec{x}) = \text{proj}_\ell(\vec{x})$ is not invertible.

5. Let ℓ be a line in $\mathbb{R}^2$ passing through the origin, parallel to the non-zero vector $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$. Let $T(\vec{x}) = \mathrm{ref}_\ell(\vec{x})$ be the transformation corresponding to the reflection of $\vec{x} \in \mathbb{R}^2$ about the line ℓ, and let

$$\boldsymbol{A} = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}, \text{ where } a^2 + b^2 = 1.$$

(a) Show that the standard matrix of $T(\vec{x})$ is given by a matrix of the form $\boldsymbol{A}$.

(b) Conversely, show that $\boldsymbol{A}$ must represent a reflection about some line. [*Hint:* Find non-zero, orthogonal vectors $\vec{v}, \vec{w}$ such that $\boldsymbol{A}\vec{v} = \vec{v}$ and $\boldsymbol{A}\vec{w} = -\vec{w}$, and show that $\boldsymbol{A}\vec{x}$ reflects $\vec{x}$ about the line $\ell = t\vec{v}$.]

(c) Determine the inverse of the matrix $\boldsymbol{A}$.

(a) We may express the reflection about $\ell = t\vec{v}$ in terms of the orthogonal projection:

$$\mathrm{ref}_\ell(\vec{x}) = 2\mathrm{proj}_\ell(\vec{x}) - \vec{x} = 2\left(\frac{\vec{x}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}\right)\vec{v} - \vec{x}.$$

We express this in terms of the vectors' components:

$$\begin{aligned}
\mathrm{ref}_\ell(\vec{x}) &= 2\left(\frac{xv_1 + yv_2}{v_1^2 + v_2^2}\right)\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} - \begin{pmatrix} x \\ y \end{pmatrix} \\
&= \frac{1}{v_1^2 + v_2^2}\begin{pmatrix} \left(2v_1^2\right)x + \left(2v_1v_2\right)y \\ \left(2v_1v_2\right)x + \left(2v_2^2\right)y \end{pmatrix} - \begin{pmatrix} x \\ y \end{pmatrix} \\
&= \frac{1}{v_1^2 + v_2^2}\begin{pmatrix} \left(2v_1^2\right)x + \left(2v_1v_2\right)y - x\left(v_1^2 + v_2^2\right) \\ \left(2v_1v_2\right)x + \left(2v_2^2\right)y - y\left(v_1^2 + v_2^2\right) \end{pmatrix} \\
&= \frac{1}{v_1^2 + v_2^2}\begin{pmatrix} \left(v_1^2 - v_2^2\right)x + \left(2v_1v_2\right)y \\ \left(2v_1v_2\right)x + \left(v_2^2 - v_1^2\right)y \end{pmatrix} \\
&= \frac{1}{v_1^2 + v_2^2}\begin{pmatrix} v_1^2 - v_2^2 & 2v_1v_2 \\ 2v_1v_2 & v_2^2 - v_1^2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}.
\end{aligned}$$

Finally, observe that

$$\begin{aligned}
\frac{\left(v_1^2 - v_2^2\right)^2}{\left(v_1^2 + v_2^2\right)^2} + \frac{\left(2v_1v_2\right)^2}{\left(v_1^2 + v_2^2\right)^2} &= \frac{v_1^4 - 2v_1^2v_2^2 + v_2^4 + 4v_1^2v_2^2}{\left(v_1^2 + v_2^2\right)^2} \\
&= \frac{v_1^4 + 2v_1^2v_2^2 + v_2^4}{\left(v_1^2 + v_2^2\right)^2} = \frac{\left(v_1^2 + v_2^2\right)^2}{\left(v_1^2 + v_2^2\right)^2} = 1.
\end{aligned}$$

Thus, the reflection matrix is of the desired form.

(b) We wish to find a vector $\vec{v}$ such that $\boldsymbol{A}\vec{v} = \vec{v}$:

$$\begin{pmatrix} a & b \\ b & -a \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \implies \begin{cases} ax + by &= x \\ bx - ay &= y \end{cases}.$$

We require that

$$(1 - a)x = by, \quad bx = (a + 1)y.$$

Since $a^2 + b^2 = 1$, we have $b^2 = 1 - a^2 = (1-a)(1+a)$, and so the first equation gives us

$$(1-a)\,x \cdot b = b^2 y = (1+a)(1-a)\,y \quad \Longrightarrow \quad bx = (1+a)\,y,$$

which is precisely the second equation. One such solution to this equation is given by $x = b$ and $y = 1 - a$, and so we may take

$$\vec{v} = \begin{pmatrix} b \\ 1-a \end{pmatrix}.$$

We now find a vector $\vec{w}$ such that $\boldsymbol{A}\vec{w} = -\vec{w}$:

$$\begin{pmatrix} a & b \\ b & -a \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix} \quad \Longrightarrow \quad \begin{cases} ax + by & = -x \\ bx - ay & = -y \end{cases}.$$

We require that

$$(a+1)\,x = -by, \quad bx = (a-1)\,y.$$

Solving these equations as before, we may take $x = a - 1$ and $y = b$, and so

$$\vec{w} = \begin{pmatrix} a-1 \\ b \end{pmatrix}.$$

Observe that

$$\vec{v} \cdot \vec{w} = \begin{pmatrix} b \\ 1-a \end{pmatrix} \cdot \begin{pmatrix} a-1 \\ b \end{pmatrix} = b\,(a-1) + (1-a)\,b = 0,$$

and indeed the vectors are perpendicular.
Given an arbitrary vector $\vec{x} \in \mathbb{R}^2$, we may thus decompose it into

$$\vec{x} = \vec{x}^{\parallel} + \vec{x}^{\perp}, \text{ where } \vec{x}^{\parallel} = \alpha\vec{v} \text{ and } \vec{x}^{\perp} = \beta\vec{w}.$$

Under the transformation $T(\vec{x})$, we have

$$\begin{aligned} T(\vec{x}) &= \boldsymbol{A}\vec{x} = \boldsymbol{A}\left(\vec{x}^{\parallel} + \vec{x}^{\perp}\right) = \boldsymbol{A}\left(\alpha\vec{v} + \beta\vec{w}\right) \\ &= \alpha\boldsymbol{A}\vec{v} + \beta\boldsymbol{A}\vec{w} = \alpha\vec{v} - \beta\vec{w} = \vec{x}^{\parallel} - \vec{x}^{\perp}. \end{aligned}$$

This is precisely a reflection about the line through the origin parallel to $\ell = t\vec{v}$.

(c) By the formula for the inverse of a 2×2 matrix,

$$\boldsymbol{A}^{-1} = \frac{1}{-a^2 - b^2}\begin{pmatrix} -a & -b \\ -b & a \end{pmatrix} = \frac{1}{-1}\begin{pmatrix} -a & -b \\ -b & a \end{pmatrix} = \begin{pmatrix} a & b \\ b & -a \end{pmatrix} = \boldsymbol{A}.$$

We thus find that $\boldsymbol{A} = \boldsymbol{A}^{-1}$: indeed, a reflection is its own inverse transformation.

6. The standard matrix of a rotation by a fixed angle α in the counter-clockwise direction is given by

$$\boldsymbol{R}_\alpha = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}.$$

(a) Let $\vec{v} = \begin{pmatrix} \cos\beta \\ \sin\beta \end{pmatrix}$. Geometrically, explain why

$$\boldsymbol{R}_\alpha \vec{v} = \begin{pmatrix} \cos(\alpha+\beta) \\ \sin(\alpha+\beta) \end{pmatrix}.$$

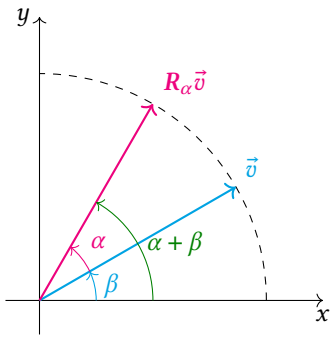
Hence, show that

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

(b) Compute the matrix products $\boldsymbol{R}_\alpha\boldsymbol{R}_\beta$ and $\boldsymbol{R}_\beta\boldsymbol{R}_\alpha$ for some angles α, β. Give a geometric description of the transformation that these products represent.

(c) Determine the inverse of $\boldsymbol{R}_\theta$, and give a geometric description of the linear transformation with standard matrix $\boldsymbol{R}_\theta^{-1}$.

(a) The vector $\vec{v}$ is the point on the unit circle whose position vector makes an angle of β with respect to the positive x-axis. Since the matrix $\boldsymbol{R}_\alpha$ rotates this vector by the angle α, the resulting vector $\boldsymbol{R}_\alpha\vec{v}$ must be the point on the unit circle whose position vector makes an angle of $\alpha+\beta$ with respect to the positive x-axis.



We compute the matrix product:

$$\boldsymbol{R}_\alpha\vec{v} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}\begin{pmatrix} \cos\beta \\ \sin\beta \end{pmatrix} = \begin{pmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta \\ \sin\alpha\cos\beta + \cos\alpha\sin\beta \end{pmatrix}.$$

We thus conclude that

$$\boldsymbol{R}_\alpha\vec{v} = \begin{pmatrix} \cos(\alpha+\beta) \\ \sin(\alpha+\beta) \end{pmatrix} = \begin{pmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta \\ \sin\alpha\cos\beta + \cos\alpha\sin\beta \end{pmatrix}.$$

(b) We compute

$$\boldsymbol{R}_\alpha\boldsymbol{R}_\beta = \begin{pmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta & -\cos\alpha\sin\beta - \sin\alpha\cos\beta \\ \sin\alpha\cos\beta + \cos\alpha\sin\beta & -\sin\alpha\sin\beta + \cos\alpha\cos\beta \end{pmatrix},$$

$$\boldsymbol{R}_\beta\boldsymbol{R}_\alpha = \begin{pmatrix} \cos\beta\cos\alpha - \sin\beta\sin\alpha & -\cos\beta\sin\alpha - \sin\beta\cos\alpha \\ \sin\beta\cos\alpha + \cos\beta\sin\alpha & -\sin\beta\sin\alpha + \cos\beta\cos\alpha \end{pmatrix}.$$

By the formulae from the previous part, we find that

$$R_\alpha R_\beta = R_\beta R_\alpha = \begin{pmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) \\ \sin(\alpha+\beta) & \cos(\alpha+\beta) \end{pmatrix} = R_{\alpha+\beta}.$$

In particular, $R_\alpha R_\beta$ and $R_\beta R_\alpha$ both correspond to the same transformation of a counterclockwise rotation by the angle $\alpha+\beta$.

(c) By the formula for the inverse of a 2×2 matrix, we have

$$R_\theta^{-1} = \frac{1}{\cos^2\theta - (-\sin^2\theta)} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$
$$\begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{pmatrix} = R_{-\theta}.$$

This matrix represents a *clockwise* rotation by the angle θ.

7. For each of the following sequences of transformations in $\mathbb{R}^2$: (i) find the standard matrix of the composite transformation, and (ii) find the standard matrix of the composite transformation when the transformations are performed in the *reverse* order.

(a) A rotation by 90°, followed by a reflection about the line $y = x$.

(b) A scaling by a factor of 2, followed by a rotation by 45°, followed by a reflection about the y-axis.

(a) The standard matrix for each transformation is

$$R_{90^\circ} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \mathrm{ref}_{y=x} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The standard matrix of the composite transformation is thus

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

In particular, this corresponds to a reflection by the x-axis. The standard matrix of the transformation performed in the reverse order is

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

This corresponds to a reflection about the y-axis.

(b) The standard matrix for each transformation is

$$S_2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad R_{45^\circ} = \begin{pmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{pmatrix},, \quad \mathrm{ref}_{x=0} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The standard matrix of the composite transformation is thus

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} -\sqrt{2} & \sqrt{2} \\ \sqrt{2} & \sqrt{2} \end{pmatrix}.$$

The standard matrix of the transformation performed in the reverse order is

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{pmatrix}\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -\sqrt{2} & -\sqrt{2} \\ -\sqrt{2} & \sqrt{2} \end{pmatrix}.$$

8. Let $\boldsymbol{A}$ be an $n \times n$ matrix.

   (a) Show that if $\boldsymbol{A}^2 = \boldsymbol{0}$, then $\boldsymbol{I} - \boldsymbol{A}$ is invertible and $(\boldsymbol{I} - \boldsymbol{A})^{-1} = \boldsymbol{I} + \boldsymbol{A}$.

   (b) Show that if $\boldsymbol{A}^3 = \boldsymbol{0}$, then $\boldsymbol{I} - \boldsymbol{A}$ is invertible and

$$(\boldsymbol{I} - \boldsymbol{A})^{-1} = \boldsymbol{I} + \boldsymbol{A} + \boldsymbol{A}^2.$$

   (c) If $\boldsymbol{A}^n = \boldsymbol{0}$ for some $n \geq 4$, is $\boldsymbol{I} - \boldsymbol{A}$ invertible?

(a) Suppose that $\boldsymbol{A}^2 = \boldsymbol{0}$. Then,

$$(\boldsymbol{I} - \boldsymbol{A})(\boldsymbol{I} + \boldsymbol{A}) = (\boldsymbol{I} - \boldsymbol{A})\boldsymbol{I} + (\boldsymbol{I} - \boldsymbol{A})\boldsymbol{A} = \boldsymbol{I} - \boldsymbol{A} + \boldsymbol{A} - \boldsymbol{A}^2 = \boldsymbol{I}.$$

(b) Suppose that $\boldsymbol{A}^3 = \boldsymbol{0}$. Then,

$$(\boldsymbol{I} - \boldsymbol{A})\left(\boldsymbol{I} + \boldsymbol{A} + \boldsymbol{A}^2\right) = \boldsymbol{I} + \boldsymbol{A} + \boldsymbol{A}^2 - \boldsymbol{A} - \boldsymbol{A}^2 - \boldsymbol{A}^3 = \boldsymbol{I}.$$

(c) From the previous parts, we might conjecture that given $\boldsymbol{A}^n = \boldsymbol{0}$,

$$(\boldsymbol{I} - \boldsymbol{A})^{-1} = \boldsymbol{I} + \boldsymbol{A} + \boldsymbol{A}^2 + \cdots + \boldsymbol{A}^{n-1}.$$

Then,

$$\begin{aligned}(\boldsymbol{I} - \boldsymbol{A})\left(\boldsymbol{I} + \boldsymbol{A} + \cdots + \boldsymbol{A}^{n-1}\right) &= \left(\boldsymbol{I} + \boldsymbol{A} + \cdots + \boldsymbol{A}^{n-1}\right) - \left(\boldsymbol{A} + \boldsymbol{A}^2 + \cdots + \boldsymbol{A}^n\right) \\ &= \boldsymbol{I} + \boldsymbol{A} - \boldsymbol{A} + \cdots + \boldsymbol{A}^{n-1} - \boldsymbol{A}^{n-1} + \boldsymbol{A}^n = \boldsymbol{I}.\end{aligned}$$

Note that this is similar to the formula of a finite geometric series with first term 1 and common ratio x:

$$(1 - x)^{-1} = 1 + x + x^2 + \cdots + x^n.$$